What should we teach our (undergraduate) students?

Terence Tao · 8 November 2026

https://terrytao.wordpress.com/2026/11/08/what-should-we-teach-our-undergraduate-students/

tao2typ · converted to Typst 2026-11-10 (UTC)

[This is a guest post by Jana Gevertz and Steffen Marcus. This blog post was initially written in a different file format and converted using AI. — T.]

In the last few months AI has hit our discipline with a frightening force. Many of us, understandably, have turned our attention to our students. They are absorbing this change while still working out what kind of mathematicians, and what kind of professionals, they hope to become. On this blog, Álvaro Lozano-Robledo recently asked "what should we tell our students?" (What should we tell our students? | What's new). In this post, we will focus on the natural follow-up question: "what should we teach them?"

Whatever our students choose for their future careers, be it academia or industry, their work will continue to depend on the depth of their mathematical understanding. However, more and more of that work will likely involve prompting, interpreting, checking, and communicating mathematical ideas that an AI model helped generate. Mathematical understanding as we have come to teach it is built on a time-honored tradition of exercise, proof, and computation: struggle with hard problems until a discovery is made, develop a conjecture and make an effort to prove it, compute by hand to uncover a desired result or a pattern that reveals itself. How will our students build mathematical understanding when AI tools can automate the very processes needed to develop it? What does it mean to train new mathematicians?

Presuming society continues to cede more decisions and reasoning to AI, it has been rightly argued that it would be dangerous for these decisions to "rest on reasons that no human community understands" (We're gonna need a lot more mathematicians | What's new). In mathematics, as in many other academic communities, understanding begins with undergraduate training that builds off years of education at the primary and secondary level. Without thriving undergraduate mathematics programs, the mathematics community is at risk of depleting the pool of individuals with the capacity to assess work done with (or by) AI and to set the directions of the field.

However, undergraduate education is at an impasse. In our liberal arts style college here in New Jersey, we have struggled with the practical implications of AI on our classrooms while circling more broadly around the question of how our curriculum must change for the future. There is a visceral feeling that our courses need to change, but it's not clear what that change should look like. The discipline's conversations so far have centered on our research and publications, our early career colleagues, and our graduate students. The 2026 AMS Committee on Education Workshop Report (2026 AMS Committee on Education Workshop Report | What's new) highlights the beginning of our efforts to reorganize graduate level curricula with AI

in mind. We believe a parallel and continuous effort focused on our undergraduate programs is needed.

This is especially important for our students entering graduate school, as they will be interfacing with mathematics differently than we have in the past. If it is true that "we won't get to be heroes anymore" (When AI comes for Your Purpose in Life – The New York Times) and we will begin working in larger teams along with AI systems to develop new ideas and proofs, then our training needs to be focused on understanding and robustly checking results presented to us in slightly "alien" forms.

The central problem: teaching understanding

Luckily, this is not really a completely new way of thinking and teaching. After all, undergraduate training has always been about teaching students to check, interpret, and explain results already known to other mathematicians. Whether it be calculations that have been executed time and time again, or proofs of important theorems that are well-established, undergraduate students learn mathematics and shape their understanding using pre-existing knowledge. From that perspective, the basic skills used to interpret, understand, and communicate mathematics learned through an AI system are already a central piece of undergraduate curricula.

However, we cannot simply continue to ignore this technology as a nuisance to our ability to assign homework exercises and hope for the best. We must teach students to work with these systems. And we must find the right way to do so without offloading their ability to think through problems on their own. Our students need both the skillset and the experience necessary to critically validate AI outputs. Whether we like it or not, they will be expected to work in this way with AI systems in their careers. Our coursework needs to carefully train them to develop mathematical intuition, reasoning, and computational skills in concert with this new technology, rather than apart from it, as is our current instinct.

Our pre-AI experience shows that there are distinct differences between productively struggling to re-create existing knowledge and using technology to do this work. Consider an undergraduate student learning strategies of integration in a calculus class. From the student's perspective, there is something distinctly different about identifying the substitution that transforms a seemingly "impossible" integral into an easily solvable one versus reading through a solution where the substitution has been selected. A student who considered the structure of the integrand to identify an appropriate substitution would indeed have created the result for themselves, despite the expected substitution being well known. A student who instead blindly followed a computer's (or even their instructor's) approach without investigating will likely not internalize that changing variables is a foundational mathematical technique that maps difficult problems onto solvable problems.

This is a small example of the difference between teaching the concept versus teaching the computation. This battle has been playing out in our discussions of pedagogy and content since symbolic computation software have become widespread. What has changed is that this conflict has now expanded to the advanced levels of our undergraduate curricula — the newest AI models have come to make our courses a push-button exercise. If our students rely on reading out computer generated proofs and computations exercise by exercise, they immediately lose the critical moments of productive struggle necessary to build mathematical understanding.

Where do we go from here?

As with any debate, there are those who will passionately argue both extremes. Some will say we simply must eschew AI technology by continuing to teach as we always have, relying on in-class pen and paper assessments. This approach fails to prepare our students for a future in which they will need to work efficiently with AI models and manage their output. Others will argue we should throw away our outdated curricula and embrace the transformative power of AI. The risk in this approach is the outsourcing of student learning and understanding, a risk that is simply too high. The answer must lie somewhere subtly in the middle of these two extremes.

In her recent post on this blog (AI, Mathematics, Education, and a public goods problem | What's New), Gireeja Ranade argues that "we must teach harder, better, and stronger". Ranade identifies the central issue facing our community as we learn how to teach our students to work with AI models: "we need to train students so that they know what they know and don't know — students who can identify proof by intimidation and by authority, and have the confidence to push back." Looking around, it seems the natural inclination for many of us is to revert to how we have taught in the past (perhaps "harder, better, and stronger") and to ignore the power of AI within the confines of the classroom. Most of us are quietly choosing one of the two extremes.

Indeed, as authors of this post ourselves, we have yet to find the right way to incorporate AI in our mathematical biology or geometry classes this semester, though we have been thinking about it for a while. Adjusting an undergraduate curriculum is difficult and time-consuming work that cannot simply be done on top of the full-time job faculty already do. Reaching a consensus is also going to be very challenging, and even once a curriculum is agreed upon, the work that must follow to redesign course materials is substantial. For these reasons, it is easy to let our undergraduate courses wait three, four, or five more years until we see how everything shakes out. However, our students desperately need us to find this delicate balance more quickly. They need to acquire a breadth of experience interfacing with these systems in their mathematical work now, or they risk getting left behind.

And therein lies the challenge: how do we design undergraduate mathematics curricula that takes such a balanced approach? There have been a number of recent summits, reports, and statements regarding how AI is changing the mathematical landscape and how we should respond. Similar concerted efforts with a focus on undergraduate education are necessary. The MAA Statement on AI in Higher Education outlines some top-down beliefs on how the undergraduate curriculum may have to change. However, much more activity is needed to help guide department leaders and educators on how to make these suggestions concrete. Together we need to work towards a modernized curriculum that considers the following questions.

- How do we design and implement a curriculum that places a greater emphasis on higher-order competencies, including conceptual and computational reasoning as well as problem solving and modeling (per: MAA Statement on AI)? Efforts to change curricula and teaching in similar ways have often been met with hesitation.
- How do we incorporate the critical evaluation of AI output into our courses so that students can robustly check these results?
- Should we be incorporating formalization techniques such as Lean into our upper-level and proof-based courses?

- Can we reduce the number of topics within each course to create space to focus on reading and interpreting AI output and comparing the answers it gives to the theory students understand to be true?
- Can we reduce the focus on arduous computations while continuing to emphasize the procedural fluency necessary to critically analyze mathematical arguments?
- How do engineering, science, social science, business, and health science disciplines see their mathematics, statistics, and data science educational needs changing in the coming years? We certainly must collaborate with related disciplines to thoughtfully consider how moving away from computation to concept and interpretation impacts the training of their students.
- How do we create courses that emphasize humanized mathematical and statistical communication?

This list is in no way exhaustive, but we do hope it can jumpstart the conversation.

Faculty in all departments need to be thinking hard about how our curriculum should change, before external players force change upon us. Continued leadership from mathematical organizations, including the AMS and MAA, as well as private organizations, is needed. The federal government would also be well-served by offering grants to support the development of innovative mathematics curricula for the AI era. After all, AI technology does not exist without the work of mathematicians. Together, we can work to give our students what they are owed: an education that prepares them to enter a world where they can effectively use AI as a tool, but not as a crutch.

Comments (9)



Anonymous · 8 November 2026

Why not be conservative? It is really not that difficult to use AI. Only the industry tells people that you'll be left behind if you don't use AI.

In reality it takes two days to surrender your mind to the machine. There is no need to teach it at all apart from attracting industry sponsors.



Kęstutis · 8 November 2026

The mindless need to "accelerate" at a time when the anthroocene as a civilization crisis has become undeniable must be recognized as a cancer within in the whig spirit which shapes many institutions itself. Fermat supposedly proved his famous last theorem without any fancy technology pertaining to modular elliptic curves. How many people have actually read older greek or islamicate text about geometry? Telling undergraduate students the only viable path is asking your questions to the Oracle machine is a bad 1970s sci fi plot..

Well frankly it does seem we live in Metal Gear already, a world where we're all hypnotized with data and numbers and it is herecy to beileve fallable puny humans are even capable of processing any of that. The climate is fine, the oracle machine says so, rising co2 emissions are of no concern is the conclusion it's analysis of the data. Everything is controlled and optimized, and our religion is the worship of Henry Ford 's assembly line, this observersation of our Brave New World is probably already a stale reaction to the times in itself.

> I really don't feel I'm exaggerating at all in a world where Nature has burnt their reputation on Microsoft's questionable qubits or publishing claims of room temperature superconductivity and the moniker for big budget science "Manhattan Project" is thrown around constantly to refer to the infrastructure buildout for "AI". This sin is really within the modern culture of Mathematics itself, indeed the Simons foundation seemingly sort of owns Arxiv now, the technology he famously built certainly has a wonderful legacy for our ability to democratically participate in our governments so huge wall street firms don't destroy our backyards with toxic chemicals…
>
> Many categorist theorist predictably took the opportunity of the chaos and confusion to give a sales pitch for their preferred specialty which of course draws attention to Grothendieck's prophetic moral stand or self aggrandized schizotypal breakdown, the accounts differ rather extremely indeed, around the time of Mai 1968, the Vietnam war and the counter culture.
>
> Here in America, if one wants to emphasize and preserve the humanity of their math departments, they need to confront your superiors who violently crushed your students voice of outrage about genocide and mass surveillance enabled by this technology. It's exactly the same picture as policy changes like the "genesis mission" mandating AI usage in order for research to recieve grants. Though I am totally with Terrance Tao trying to avoid politics as much as he can, the ideology of the times went for him anyway, and I know first hand his declaration of math not being political mattered absolutely not to supporters of that iseology who lable him the other and enemy anyway. Pretending we do not actually have to directly confront that influence upon policy is an shambolic farce, unless one is content for the future of math to be small little Pythagorean cults in a world resembling Wall-E…



Ben Johnsrude · 8 November 2026

To the bullet point "[s]hould we be incorporating formalization techniques such as Lean into our upper-level and proof-based courses," it seems to me that the answer is *no*, at least if "incorporation" means teaching students that a solution has been found iff a Lean file has been produced.

The trade-off with using formal proof systems is that we gain certainty in the truth of the output at the cost of other desiderata; a student can arrive at a solution to the problem "prove that addition is commutative" in the natural number game without ever having a coherent idea of what the strategy of the proof was. In an extreme case, a Lean file proving X can spend arbitrarily many lines of code proving unrelated facts; so, a student learning the proof of the inverse function theorem can come to believe that it involves a lot of group theory. The example's a bit silly; but, I think anyone who has played the natural number game can sympathize with the experience that a lot of it is blundering in the dark until the green light goes off, without having acquired any mathematical understanding.

The point, of course, is to indicate that the production of a Lean file should not be treated by students as sufficient for having solved a problem. Some sort of understanding beyond this is necessary.

 **akkapi** · 8 November 2026

Are instructors necessary to teach the youth how to interact with AI to get answers? I wonder how well the current generation of mathematicians itself knows how to best use AI for research, versus the variety of better and more optimal paths that the younger generation will discover and could teach the mathematicians of today.

The point of mathematical instruction by mathematicians should be to teach mathematics, rather than bifurcate tasks both for the instructors and the students. Considering that there is so much worry about students using AI to do their homework, probably spending more instruction time on this is less useful than accepting that this would happen anyways and focusing on assessing how much the students have learned *even with* the help of AI. In this sense, the focus might shift to *how to assess students better* "in the age of AI". Improvements in this regard, without increasing stress both for the students and the instructional team (the teachers, the graders, etc.) will likely go much farther: good strategies would implicitly enforce students to be more critical with regards to what they may read from AI output.

Teaching formalization *tools* like Lean seems to also put the cart before the horse. How well do mathematicians in fields outside of the nucleus of formalization (type theory, calculus of inductive constructions, etc.) actually understand the theory of Lean or can explain why it works? Should emphasis be placed in mathematical instruction on learning tools regarding proof that the instructor themself doesn't fully understand? A key issue in "trusting AI" is the interaction between the natural language and Lean outputs it produces, which are only very recently being properly studied in research mathematics: advancements in this direction are likely good candidates for future special-topics courses. As has been done with other areas of mathematics, it seems there would be more to be gained by offering better courses and putting some emphasis on the foundations of these tools, in addition to the tools themselves.

Ultimately, what is the point of teaching mathematics? At the risk of oversimplification, a basic analogy is that although there are many tricks in abstract algebra, a good group theory course still requires the student to go through computations by hand to get a feel for how symmetries behave based on structure. This approach of doing things "from first principles" is present in all parts of existing mathematical instruction, where what constitutes the "first principles" moves as one advances along the theory. In mathematics as a whole, however, the development of proof, or a tight logical argument from start to finish, is by definition "the" first principle. If the point of teaching mathematics is for the student to learn mathematics, then it seems that teaching how to instruct AI to do mathematics or to "think in the Lean language" (or similar) would be a distraction, both for instructors as well as students.

 **sma045** · 8 November 2026

My father was a Physics-trained Electrical Engineer. When the subject of rotating something by 90 degrees arose in real world contexts, he would say "multiply by i". Geometrical understanding is crucial, and increasingly that is in more than 3 dimensions. The route to such understanding lies through Real Clifford Algebra that is increasingly called Geometric Algebra (GA), where we find many things that square to -1. This creates many situations where there seem to be these mysterious imaginary numbers. Introducing them makes the mathematics simpler, while losing the geometric understanding. David Cahoe

(sudgylacamore) has educational youtube videos on GA, and he recently gave a talk at AGACSE2026 (https://youtu.be/R5Nd2VmF0-M?si=BpGyIBpBL349-Z5l) on how to teach it, which he is in the process of implementing in his Zero to Geo videos. The talk details the way we are easily seduced into explanatory paths that are ultimately misleading.

Algebraic Integration is inverse differentiation. It is just one of many cases where finding inverses is hard. Is it important for actual practical applications of Mathematics, where we are mostly forced into approximate computational methods? Or is it a distraction from getting the geometrical understanding? What it is, is a good way to get students to think clearly when dealing with symbolic algebra. A productive, though somewhat brutal setting for this is competition. Just as chess learnt how to continue, I think the problem solving stuff that builds intuition can be continued through competitive activities at various levels. While this can be just pen and paper tests and exams, it can also be teams, including teams with equal access to local AI. And lets build it all the way up to an expanded Mathematics Olympiad.

 **Paul Leopardi** · 8 November 2026

I would be very interested to hear your views on service teaching of mathematics for non-mathematicians who will be using mathematics-based techniques throughout their careers, How should we be now teachingour students about how to apply mathematical models and methods? How to assess, adapt, adopt, translate into specialized aplication domains? Why should mathematicians be the ones to undertake such teaching?

 **Anonymous** · 8 November 2026

“From the student’s perspective, there is something distinctly different about identifying the substitution that transforms a seemingly “impossible” integral into an easily solvable one versus reading through a solution where the substitution has been selected”

This brings to mind the difference between a natural birth, in which there is struggle and accomplishment, and birth by C-section in which there is neither. At what point does a student give up trying to solve an assigned problem and resort to AI? Somehow I doubt there will ever be an effective remedy for this problem.

 **Anonymous** · 9 November 2026

Yet another “understanding” post without any analysis of what it is or why it is needed (or not).

> Can we reduce the number of topics within each course to create space to focus on reading and interpreting AI output and comparing the answers it gives to the theory students understand to be true?

No! Courses that involve writing proofs already target those skills. Enhance and extend those.

> Should we be incorporating formalization techniques such as Lean into our upper-level and proof-based courses?

No. Same as reason as above. There could perfectly be elective undergraduate or graduate courses specifically about formal systems or specific ones. Proof-based courses already have plenty on their plate with studying general patterns on proof techniques (induction, contradiction, contraposition, infinite descent, extremal elements, …)

> How do we create courses that emphasize humanized mathematical and statistical communication?

Ewww! “humanized mathematical [] communication”. I once saw a school program in mathematics which first goal was to “understand our feelings about mathematics”. What?

Some educators have never learned the lesson that teaching objectives have to have direct translation into evaluation. They must be measurable. “Understanding polynomial equations” is not a teaching objective. “Determining properties of the roots, like sign, number, or location, from the coefficients of the polynomial equation” is a teaching objective.

Instead of some hippie idea like “humanized mathematical communication” that makes no sense, use the time to teach the correspondence between common language expressions and algebraic or arithmetic ones: “for each” leads to multiplication, “it is possible” to “exists”, “impossible” to “for all”, “increased by” to addition, that “any” shouldn’t be used in mathematics because it is ambiguous and can be read as both “exists” or “every”, etc. I used to study from an algebra book, from a time way before my time, that had an entire chapter dedicated to such translations. Ultimately, the skill of such language translation is what you really have in mind, and what constitutes a measurable objective.



Anonymous · 9 November 2026

I think you have misinterpreted our meaning of “humanize”. We mean that humans need to be able to understand the proof. A proof a machine could read, but not a human, is not our goal.